Boat power findings

Race tracker boat unit (Raspberry Pi 3A+ with ArduSimple simpleRTK2B and XBee) · measured 13 Oct 2026 with an FNIRSI FNB58

1. What was measured

Run	State	Samples	Average	Min / max	Current avg / peak
run1	Boat (Pi 3A+ and ArduSimple) as running	2,232 (22 s)	1.73 W	1.60 / 2.32 W	0.341 / 0.457 A
run2	Same state, repeat	3,000 (30 s)	1.73 W	1.58 / 2.41 W	0.341 / 0.474 A
run3-noradio	Pi 3A+ only, ArduSimple USB unplugged	2,732 (27 s)	0.94 W	0.60 / 2.00 W	0.184 / 0.394 A

- Supply was a steady 5.08 V. Peak current is under 0.5 A, far below the 2.5–3 A limit of the UPS boards considered.
- The 4 W bursts seen on the meter display do not appear in any log. Nothing exceeded 2.5 W. Each run is only 20–30 s, so a burst every minute or so would be missed.
- run3 is noisy (1-second averages 0.64–1.19 W), so treat 0.94 W as approximate.

2. Where the power goes

Part	Power	Share	Basis
Pi 3A+	about 0.94 W	54%	run3 (normal for a 3A+ idle, from memory about 0.8–1 W)
ArduSimple board (GNSS, XBee, board)	about 0.79 W	46%	run2 minus run3
of which XBee in receive	about 0.1 W	about 6%	datasheet, 29 mA (estimate)
of which ZED-F9P chip	about 0.2 W	about 12%	datasheet, about 68 mA at 3.0 V (estimate)
of which board, antenna, USB bridge	about 0.45–0.5 W	about 28%	remainder, unexplained

The run3 drop of 0.79 W is the whole ArduSimple side (GNSS, XBee and board), not the XBee alone. The XBee's own share is small.

3. Battery runtime

Assumes a 10,000 mAh pack is 37 Wh nominal and about 30 Wh usable at 5 V (80–85% conversion). Real packs often deliver 10–20% less than the label, so plan on about 10% below these figures. The 25.9 Wh pack from the Innovisory report is taken as about 22 Wh usable.

Configuration	Draw	10,000 mAh pack	25.9 Wh pack
3A+ boat, awake (measured)	1.73 W	about 17–18 h	about 12–13 h
3A+ boat, wake-frame sleep (estimate)	about 1.4 W	about 21–22 h	about 15–16 h
3A+ Pi only (measured)	0.94 W	about 32–33 h	about 23 h
Zero 2 W boat, awake (estimate)	about 1.3–1.5 W	about 20–23 h	about 15–16 h
Zero 2 W boat, wake-frame sleep (estimate)	about 1.05–1.2 W	about 25–28 h	about 18–21 h
Zero 2 W Pi only (estimate)	about 0.55–0.7 W	about 43–55 h	about 31–40 h

Zero 2 W figures use published idle numbers (about 0.5–0.6 W), not a measurement on this boat. Whether the agent runs comfortably on a 1 GHz Zero 2 W with 512 MB, and the timing guard, have not been tested.

4. Wake-frame sleep mode

- The repo's radio sleep mode (base sends a power frame) sleeps the XBee through SLEEP_RQ and also puts the GPS in software backup (about 1.4 mA).
- Estimated whole-boat draw asleep is about 1.4 W (Pi 0.94 W plus about 0.45–0.5 W of board overhead), a saving of about 0.3 W, around 18%. That is about 4 hours on the 10,000 mAh pack. This is an estimate and needs a meter run to confirm.
- It needs one wire (GPIO17 to XBee pin 9 with a 10 k pull-down) and `ATSM1`. Whether pin 9 is reachable on the ArduSimple socket is unconfirmed.
- The GPS enters backup only if its USB port is not connected, otherwise `GPS_SLEEP_FORCE_USB=1`. Waking costs a GPS reacquisition unless V_BCKP is fitted.
- Conclusion: not worth wiring for battery life alone. Its value is operational (idle boats stop transmitting and free their slots).

5. Bigger levers

1. **Power the whole boat off between sessions.** Saves 100% while off. At 14 h off a day and 10 h on, daily use is about 17 Wh, so a 10,000 mAh pack covers about 1.7 such days. It does not add running hours, only spreads them over more calendar days unless the pack is charged overnight. The GPS cold-starts each morning unless V_BCKP is fitted.
2. **Trim the Pi (0.94 W):** Wi-Fi off during a race, HDMI and LEDs off, fewer cores or lower clock. All are cheap to test with no wiring.
3. **Zero 2 W instead of a 3A+:** maybe 0.3–0.4 W saving, about 15–20% more runtime, at the cost of a board change (micro-USB OTG port, 2.4 GHz Wi-Fi only).
4. **Find the ArduSimple's 0.45–0.5 W of overhead:** test with the antenna, XBee or USB bridge removed one at a time.

6. Power management hardware

- **PiSugar 3 (Pi Zero form factor):** 5 V at 2.5 A, 1200 mAh (about 4.4 Wh, about 1.9 h at 2 W), RTC, timing boot, soft shutdown, I2C. No external battery connector listed. Fit on a 3A+ not confirmed. Scheduled power-off and low-battery shutdown are not documented in what was read.
- **Alternatives, from memory, unverified:** Witty Pi 4 (scheduler, no battery of its own), PiJuice Zero (RTC, JST battery), Waveshare 18650 UPS modules.

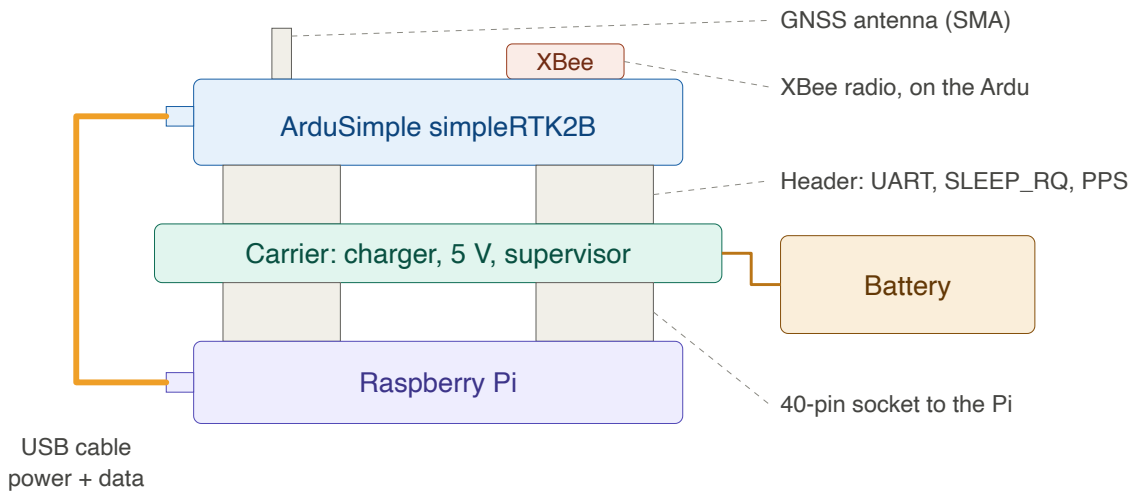
- **Scheduled shutdown already exists** in the repo (rover dashboard card, `ROVER_SHUTDOWN_AT`). It only powers down, never up, so wake hardware is still needed.
- **Waking a halted Pi:** GPIO3 pulled low wakes it from halt (from memory, unverified on a 3A+). There is no Wake-on-LAN. Waking from the XBee would need the Ardu powered while the Pi is halted and a DIO route to the Pi, which is not built.
- **Logging:** the FNB58 works over USB on this Mac with the baryluk logger (`sudo ./venv/bin/python fnirsi_logger.py > run.txt`; stop with Ctrl-C or by creating `fnirsi_stop`; remove that file before the next run).

7. Proposed next measurements

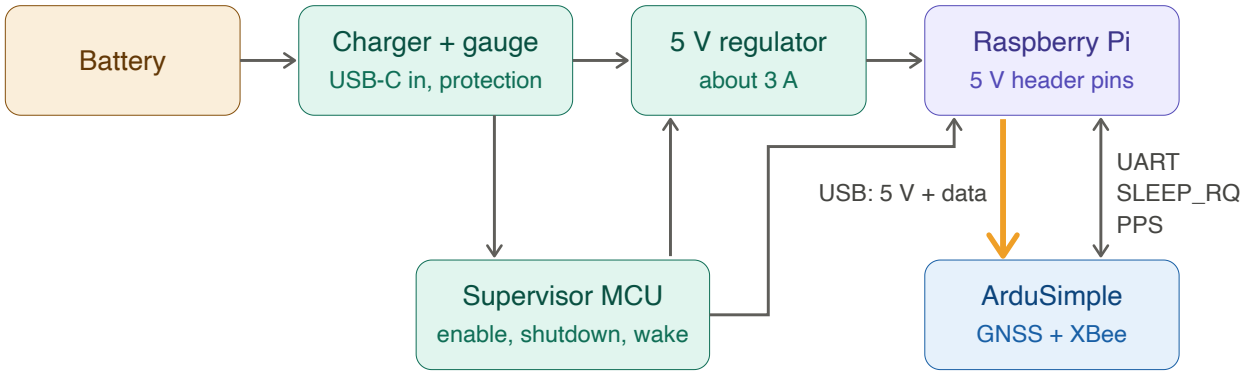
1. Pi only with the agent stopped (idle floor).
2. Boat running, 2–3 minutes, while actually transmitting (to catch the 4 W bursts).
3. The same with Wi-Fi off.
4. The same with the XBee asleep or removed.
5. A battery run-down test on the actual pack.

8. Carrier board concept (not a verified design)

The idea is a board with a 40-pin socket on its underside for the Pi and a header on top for the ArduSimple, so nothing is hand-wired. The pin mapping, V_BCKP, SLEEP_RQ and PPS access on the ArduSimple header, and the Pi's USB 5 V path current have not been checked. It was sketched for a Pi Zero form factor; the boat currently uses a 3A+, so the footprint and USB port would change.



Side view: Pi plugged into the underside, ArduSimple on top. The USB cable carries the Ardu's only power and its data.



Power and signals: battery to charger to 5 V regulator to the Pi; the Pi powers the Ardu over USB; UART, SLEEP_RQ and PPS run between them.

9. Open questions

- What pack will the boats use (chemistry, voltage, capacity, connector, power bank or bare cell)?
- How long must a boat run per charge: a race, a regatta day, practice only?
- Does the power bank stay on at light load (Pi only is about 0.18 A)?
- Do you need to start boats remotely, or is a timed or button start enough?
- Stay with the 3A+ or move to a Zero 2 W, and what does the carrier footprint then need to be?
- Where is the real 4 W burst, and does it show up in a longer transmitting run?